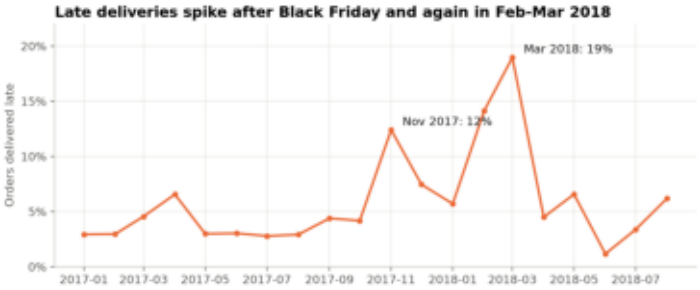
What drives bad reviews in Brazilian e-commerce?

Analysis of 96,211 delivered Olist orders, Jan 2017 – Aug 2018 · SQL, Python, Excel, Power BI

R$15.4M	96,211	R$160	6.8%	4.16
revenue	delivered orders	avg order value	delivered late	avg review (1-5)

Late delivery is the biggest driver of bad reviews





Key findings

1. Late delivery drives bad reviews: on-time orders average 4.29 stars, 4+ days late about 2. Late orders are 6.7% of reviewed orders but 32.6% of all 1-2 star reviews.
2. North-east customers get late orders up to 5x as often as São Paulo (Alagoas 21.5% vs 4.5%).
3. The late rate tripled after Black Friday 2017 (12.4%) and peaked at 19.0% in March 2018.
4. Orders split across several sellers score 1.3 stars lower (2.85 vs 4.17), though almost never late.
5. Revenue grew 2.8x from early 2017 and has held at about R$1M a month since March 2018.
6. Only 3.0% of customers ordered twice; São Paulo state brings in 37% of revenue.

Recommendations

- → Lengthen delivery estimates or change carriers for north-east routes.
- → Book extra carrier capacity before big promotions like Black Friday.
- → Warn customers about split parcels at checkout, or consolidate shipments.
- → Send a repeat-purchase voucher after every on-time delivery.

How it was built

SQL	SQLite database with keys; 15 queries using CTEs, window functions and data-quality checks.
Python	pandas data-quality pass (26 checks, 19 issues documented) and exploratory analysis notebook.
Excel	Monthly tracker: SUMIFS/XLOOKUP summaries, dropdown filters, conditional formatting, sparklines.
Power BI	Star-schema model with DAX measures for revenue, late rate, reviews and repeat customers.

Ravi Yadav · Data: Olist dataset on Kaggle

github.com/RAVIYADAV6522/olist-ecommerce-analysis